



Level 1

Challenge C



Directions:

In the `Haskell_Challenges.hs` file, create a function called `"addLists"` that does the following:

The function will take in two lists of floats and return a list of their element wise sums. It will need to add corresponding items in each list to create this list of sums. You must check to see if both lists are the same size and you should return empty list if they are not.

You will have to figure out an appropriate data type for this function and then define it in the challenges file.

Examples:

```
{- call -}  addLists [1.0, 1.5, 2.0] [1.0, 1.0, 1.0]
{- returns -}  [2.0,2.5,3.0]
```

```
{- call -}  addLists [1, 2, 3, 4, 5] [5, 4, 3, 2, 1]
{- returns -}  [6.0,6.0,6.0,6.0,6.0]
```

```
{- call -}  addLists [] []
{- returns -}  []
```

```
{- call -}  addLists [3.14] [48.1234]
{- returns -}  [51.2634]
```

```
{- call -}  addLists [1.0] [1.0, 2.0]
{- returns -}  []
```